

CHEMISTRY

1. Correct order of EN in below elements

- (a) $1s^2 2s^2 2p^4$ (b) $1s^2 2s^2 2p^3$
 (c) $1s^2 2s^2 2p^5$ (d) $1s^2 2s^2 2p^6$
 (1) $a > b > c > d$
 (2) $b > a > c > d$
 (3) $c > a > b > d$
 (4) $a > c > d > b$

Ans. (3)

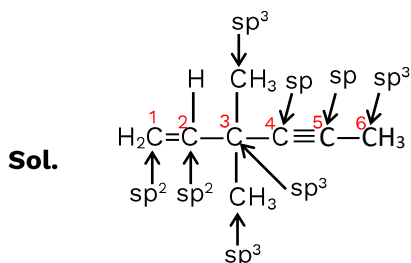
Sol. Electronic configuration:

- (a) $1s^2 2s^2 2p^4$ = Oxygen
 (b) $1s^2 2s^2 2p^3$ = Nitrogen
 (c) $1s^2 2s^2 2p^5$ = Fluorine
 (d) $1s^2 2s^2 2p^6$ = Neon
 EN order = $c > a > b > d$

2. In 3, 3-dimethylhex-1-en-4-yne, the number of sp , sp^2 and sp^3 carbon atoms, respectively are

- (1) 2, 2, 4
 (2) 2, 2, 2
 (3) 4, 2, 2
 (4) 2, 4, 2

Ans. (1)



3. Among the following molecules which one has sp^3d hybridized having lone pair and having different bond length:

XeF_2 , XeF_4 , PF_5 , SF_4

- (1) XeF_2
 (2) XeF_4
 (3) PF_5
 (4) SF_4

Ans. (4)

Sol. SF_4 , $4\sigma + 1$ l.p; sp^3d hybridisation. Axial & equatorial bond length are present in this molecule.

4. 0.1 g of organic compound was subjected to estimation of N by Duma's method. Volume of N_2 evolved at $0^\circ C$, 1 atm was 11.2 ml. Find % nitrogen in organic compound

Ans. (14)

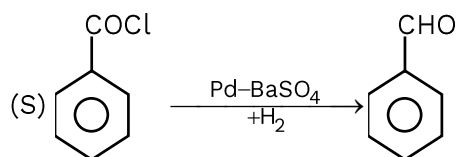
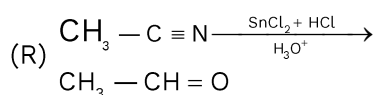
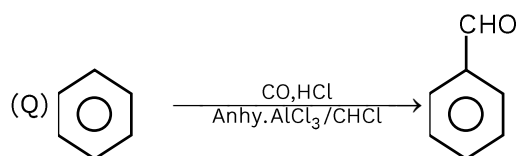
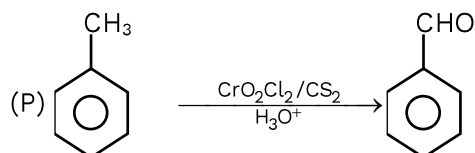
Sol. $n_{N_2} = \frac{11.2}{22400} = \frac{1}{2000} \text{ mole}$

$$\text{Mass} = \frac{1}{2000} \times 28 = 14 \times 10^{-3} \text{ gm}$$

$$\% N = \frac{14 \times 10^{-3}}{0.1} \times 100 = 14\%$$

5. Match the coloum

**Coloum-I
(Reaction)**



**Coloum-II
(Name of the reaction)**

(1) Rosenmund Reaction

(2) Etard Reaction

(3) Gatterman-Koch Reaction

(4) Stephen Reaction

(1) P→1; Q→2; R→3; S→4

(3) P→2; Q→4; R→3; S→1

(2) P→2; Q→3; R→4; S→1

(4) P→3; Q→4; R→1; S→2

Ans. (2)

Sol. Conceptual

6. Nature of compounds TeO_2 and TeH_2 is _____ and _____ respectively

(1) oxidising and reducing

(2) Highly acidic and highly basic

(3) reducing and oxidising

(4) None

Ans. (1)

Sol. Conceptual.

7. Statement-I: Melting point of neopentane is greater than that of n-pentane

Statement-II: Neopentane give only one mono-substituted product.

(1) Both Statement-I & II are correct

(2) Both Statement-I & II are Incorrect

(3) Statement-I is correct & Statement II is Incorrect

(4) Statement-I is Incorrect & Statement II is correct

Ans. (1)

Sol. Melting point depend on Intermolecular force of attraction & packing of lattice.

8. Sodium nitroprusside test is used for detection of which of the following species in organic compounds

- (1) SO_4^{2-} (2) S^{2-} (3) Na^+ (4) PO_4^{3-}

Ans. (2)

Sol. $\text{Na} + \text{S}^{2-} \longrightarrow \text{Na}_2\text{S} + \xrightarrow{\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]} \text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$
(Purple Deep Violet Colour)

9. Which of the following is the order of enthalpy of atomization of 3d-series ?

- (1) $\text{Ni} > \text{Cu} > \text{Mn} > \text{Zn}$ (2) $\text{Zn} > \text{Cu} > \text{Mn} > \text{Ni}$
(3) $\text{Cu} > \text{Mn} > \text{Ni} > \text{Zn}$ (4) $\text{Mn} > \text{Ni} > \text{Cu} > \text{Zn}$

Ans. (1)

Sol. NCERT (d&f block)

10. Following 4 amino acids: Glycine, Alanine, Leucine & Valine form tetrapeptide linkage. For this tetrapeptide linkage, the number of possible sequences (primary structures) is/are:

Ans. 24

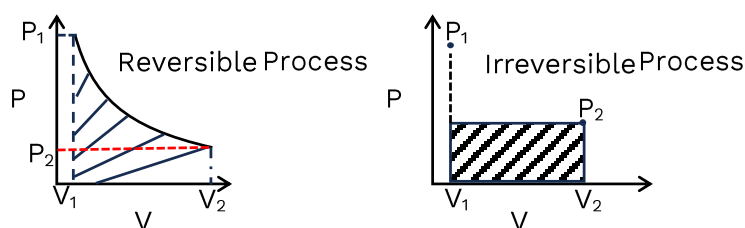
Sol. Since we have four different amino acids, we can arrange them in $4!$ ways. So number of primary structures = $4 \times 3 \times 2 \times 1 = 24$

11. In adiabatic process, the magnitude of work done in case one step & ∞ step follows order :-

- (1) $|W_{\text{rev}}|_{\text{expansion}} > |W_{\text{irr}}|_{\text{expansion}}$ (2) $|W_{\text{rev}}|_{\text{expansion}} < |W_{\text{irr}}|_{\text{expansion}}$
(3) $|W_{\text{rev}}|_{\text{expansion}} = |W_{\text{irr}}|_{\text{expansion}}$ (4) None

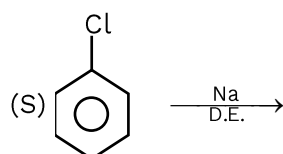
Ans. (1)

Sol.



12. List-I

- (P) $\text{RX} + \text{Na} \xrightarrow[\text{Ether}]{\text{Dry}}$
(Q) $\text{RCOOH} \xrightarrow[\Delta]{\text{NaOH} + \text{CaO}}$
(R) $\text{ROH} \xrightarrow{\text{anhy. ZnCl}_2}$



List-II

- (1) Fittig Reaction
(2) Lucas Method
(3) Wurtz Reaction
(4) Sodalime Decarboxylation Reaction

- (1) P→1; Q→4; R→2; S→3
(3) P→3; Q→2; R→4; S→1

- (2) P→3; Q→4; R→2; S→1
(4) P→1; Q→2; R→3; S→4

Ans. (2)

Sol. Name Reactions

13. List-I

- (P) Fractional Distillation
(Q) Simple Distillation
(R) Under Reduce P. Distillation
(S) Steam Distillation

List-II

- (1) Diesel + Petrol
(2) Aniline + H₂O
(3) Aniline + CHCl₃
(4) Glycerol + Pentyl

- (1) P→1; Q→3; R→4; S→2
(2) P→2; Q→3; R→4; S→2
(3) P→3; Q→2; R→4; S→1
(4) P→1; Q→2; R→3; S→4

Ans. (1)

Sol. Fact

14. The d-orbital electronic configuration of the complex among [Co(en)₃]³⁺, [CoF₆]³⁻, [Mn(H₂O)₆]²⁺ and [Zn(H₂O)₆]²⁺ that has highest CFSE is

- (1) $t_{2g}^3 e_g^2$
(2) $t_{2g}^6 e_g^4$
(3) $t_{2g}^6 e_g^0$
(4) $t_{2g}^4 e_g^2$

Ans. (3)

Sol. en(ethylenediamine) is SFL and chelating ligand.

15. For the reaction $A(g) \rightarrow B(g) + C(g)$. K_p is equilibrium and P is total equilibrium pressure, α is the degree of dissociation of A. Which of the following is correct?

- (1) If K_p is very much greater than P , α is much less than unity.
(2) If P is very much greater than K_p , α tends to unity.
(3) If P increases, α decreases.
(4) If P increases, α increases.

Ans. (3)

Sol. $A(g) \rightarrow B(g) + C(g)$

$t = 0$ 1 0 0

t_{eq} $1-\alpha$ α α

$$K_p = \frac{\alpha^2 P}{1-\alpha^2}$$

$$\text{So, } \alpha = \sqrt{\frac{K_p}{K_p + P}}$$

$P \uparrow \quad \alpha \downarrow$

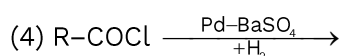
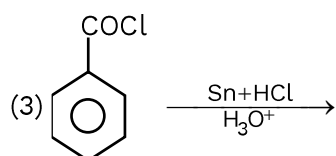
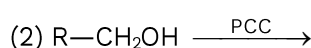
16. The number of unpaired electrons and hybridisation of $[\text{MnCl}_6]^{3-}$

- (1) $4, sp^3d^2$
 (2) $2, sp^3d^2$
 (3) $4, d^2sp^3$
 (4) $2, d^2sp^3$

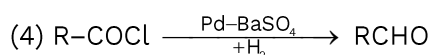
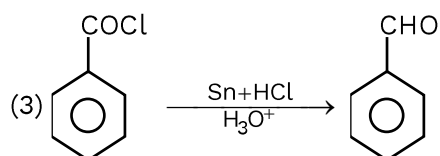
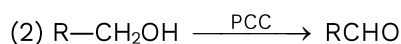
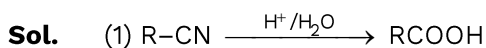
Ans. (1)

Sol. Mn^{+3} has d^4 configuration and Cl^- is a WFL.

17. Which of the following reactions give carboxylic acid as product?

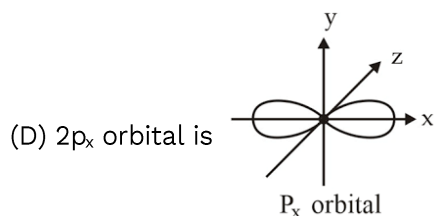


Ans. (1)



18. Consider the following statements

- (A) Value of l gives shape of orbital
 (B) Ψ represent wave function of an electron
 (C) Electron density of p_x orbital in xy plane is zero



The correct statement(s) are

- (1) (A) and (D) only
 (2) (A), (C) and (D) only
 (3) (A), (B) and (D) only
 (4) (A), (B), (C) and (D)

Ans. (3)

Sol. Theoretical